

Student's Name: _____ Teacher's Code : _____



Saint Ignatius' College, Riverview
Mathematics Assessment Task
2022

Year 12
Mathematics (Extension One)
Task 4
Trial HSC Examination
Date : 26 th August 2022

General Instructions:	Topics Examined:
• Reading time: 10 minutes • Time Allowed: 2 hours • Write using black pen • Calculators approved by NESA may be used • Attempt all questions in the booklets provided • Write your name and your teacher's code in the positions indicated • Marks may not be awarded for missing or carelessly arranged working.	Section A 10 Marks Multiple Choice Section B Short Answer Question 11 15 Marks Question 12 15 Marks Question 13 15 Marks Question 14 15 Marks Total 70 Marks
Teachers : • Mr R Maxwell REM • Mr D Reidy DPR • Mr N Mushan NHM • Mr J Newey JPN	

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SECTION I: Multiple Choice Questions:

1. Given $f(x) = \sqrt{x} - 3$, what are the domain and range of $f^{-1}(x)$?

(A) $x \geq -3, y \geq 0$

(B) $x \geq -3, y \geq -3$

(C) $x \geq 0, y \geq 0$

(D) $x \geq 0, y \geq -3$

2. What is the value of $\sin 2x$, given that $\sin x = 0.8$ and x is obtuse?

(A) $-\frac{12}{25}$

(B) $-\frac{24}{25}$

(C) $\frac{12}{25}$

(D) $\frac{24}{25}$

3. The graph of the function $y = \cos^{-1}(2x)$ is dilated horizontally by a scale factor of 4 and then translated vertically by 3 units.

What is the new equation?

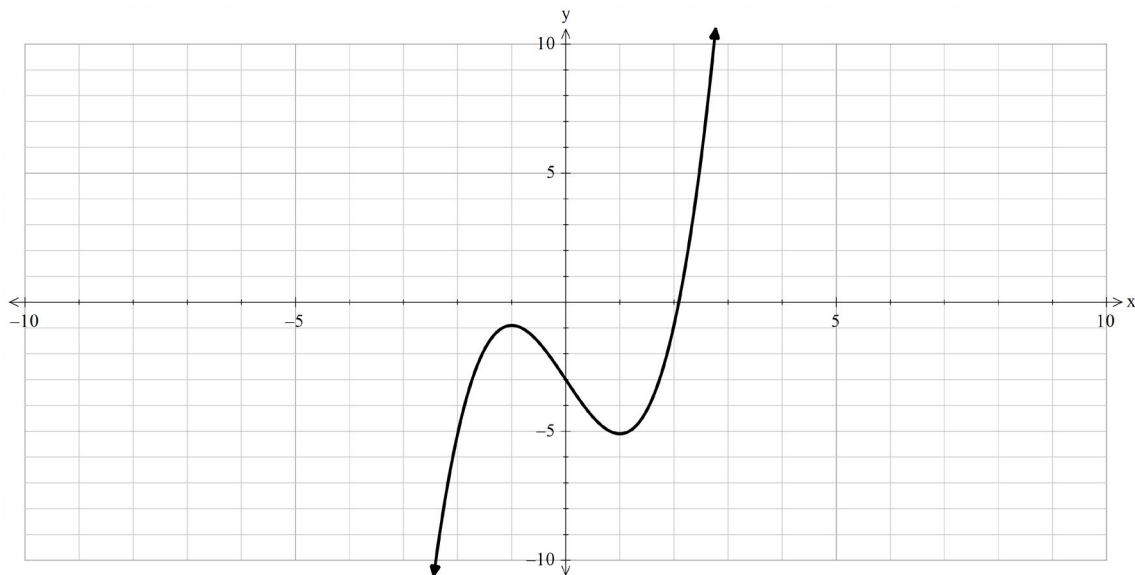
(A) $y = \cos^{-1}\left(\frac{2x}{4} + 3\right)$

(B) $y = \cos^{-1}(8x) + 3$

(C) $y = \cos^{-1}\left(\frac{2x}{4}\right) + 3$

(D) $y = \cos^{-1}(8x + 3)$

4. Consider the graph of $y = f(x)$ shown below:



Which one of the following would have 2 more roots than $f(x)$?

(A) $y = -2 \times f(x)$

(B) $y = f(x) + 3$

(C) $y = f^{-1}(x)$

(D) $y = f(x + 3)$

5. If $f(x) = \frac{3 + e^{2x}}{5}$ where $f(x) > \frac{5}{3}$ which of the following is $f^{-1}(x)$?

(A) $\ln(5x - 3)$

(B) $\frac{1}{2}\ln(5x - 3)$

(C) $\ln(5x) - \ln 3$

(D) $\frac{1}{2}\ln(5x) - \ln 3$

6. A curve is defined parametrically by $x = -\ln t$, $y = \cos 2t$ for $t > 0$.

At what approximate value of x does the curve cross the x -axis for the first time as t increases from zero?

(A) -1.7

(B) -1.37

(C) -0.86

(D) 0.24

7. Find the standard deviation of the Bernoulli random variable with the probability distribution represented by the following piecewise function.

$$P(X = x) = \begin{cases} 0.27 & x = 1 \\ 1 - p & x = 0 \end{cases}$$

(A) 0.1971

(B) 0.4440

(C) 0.03884

(D) 0.7768

8. In the cartesian plane, a vector perpendicular to the line $3x + 2y + 1 = 0$ is

(A) $3\underset{\sim}{i} + 2\underset{\sim}{j}$

(B) $-\frac{1}{2}\underset{\sim}{i} + \frac{1}{3}\underset{\sim}{j}$

(C) $2\underset{\sim}{i} - 3\underset{\sim}{j}$

(D) $\frac{1}{2}\underset{\sim}{i} - \frac{1}{3}\underset{\sim}{j}$

9. What is the simplified form of:

$$\frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}} =$$

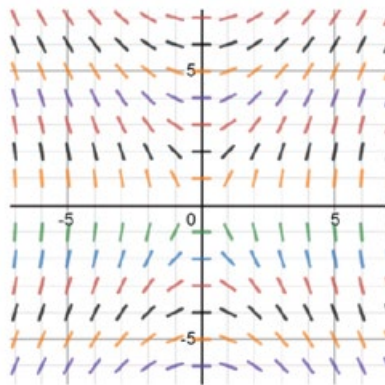
(A) $\cos \theta$

(B) $\sec \theta$

(C) $\tan \theta$

(D) $\cot \theta$

10. The slope field represent which of the following equation?



(A) $\frac{dy}{dx} = \frac{2x}{y}$

(B) $\frac{dy}{dx} = \frac{2y}{x}$

(C) $\frac{dy}{dx} = \frac{x^2}{y^2}$

(D) $\frac{dy}{dx} = \frac{y^2}{x^2}$

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SECTION II:

Question 11

Start on a new answer booklet

15 Marks

- a. A polynomial $P(x)$ when divided by $(x - 2)(x + 1)$ gives a remainder $(x - b)$. 2
When this polynomial $P(x)$ is divided by $(x + 1)$ it gives a remainder 3.
Find the value of b .

- b. A ball is rolling along a horizontal plane has position vector

$$\underline{r} = x \underline{i} + y \underline{j} \quad y \geq 0 \quad \text{and velocity vector} \quad \dot{\underline{r}} = \frac{1}{y} \underline{i} + (1 - y) \underline{j}.$$

- i. The component of velocity in the $\underline{j}$ direction gives the differential equation

$$\frac{dy}{dt} = 1 - y. \quad \text{Show that the solution to this differential equation is}$$

$$\ln|1 - y| = A - t \quad \text{Where } A \text{ is a constant.} \quad 2$$

- ii. Given that the ball is initially at the origin and that the y values are restricted to $0 \leq y < 1$. Find the equation of y in terms of e and t . 1

- iii. Hence or otherwise, find the velocity vector when $t=3$. 2

- c. Using the substitution $u = e^x + 1$ or otherwise, find the exact value 3

in a single fraction form of $\int_0^1 \frac{e^x}{(1 + e^x)^2} dx$

- d. A missile is shot from the origin O with initial speed of 64 ms^{-1} at an angle of 30° to the horizontal. The equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -10$.

- i. Show that $x = 32\sqrt{3} t$ 1

- ii. Show that $y = 32t - 5t^2$ 2

- iii. What is the cartesian equation of the trajectory of the missile? 2

Question 12**Start on a new answer booklet****15 Marks**

a. Solve $\frac{x-5}{2x+1} \geq 1$. **3**

b. Given that $y = e^{2x} + e^{-2x}$

i. Find $\frac{d^2 y}{dx^2}$ **1**

ii. Determine the values of constants a and b that satisfy the following equation:

$$\frac{d^2 y}{dx^2} + a \frac{dy}{dx} + by = 5e^{2x} + e^{-2x}$$
3

- c. A bag contains 5 identical blue marbles, 6 identical black marbles and 3 identical red marbles. Three marbles are drawn at random. Find the probability that exactly two blue marbles are drawn. (Answer in simplified fraction form) **2**

d. i. Find $\int \frac{x}{\sqrt{1-x^2}} dx$ using the substitution $u = 1 - x^2$. **2**

ii. Differentiate $x \cos^{-1} x$ with respect to x . **2**

iii. Hence, find $\int \cos^{-1} x dx$. **2**

Question 13**Start on a new answer booklet****15 Marks**

a. Given that $\underline{u} = 2\underline{i} + 3\underline{j}$ and $\underline{v} = \underline{i} + 2\underline{j}$.

i. Find $\underline{u} - \underline{v}$.

1

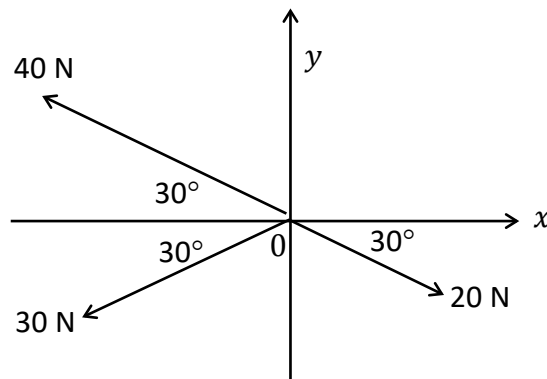
ii. If $\underline{u} - \lambda \underline{v} = -2\underline{i} - 5\underline{j}$. Find λ .

1

iii. If $a \underline{u} + b \underline{v} = \underline{i}$. Find a and b .

1

b. The diagram shows three forces acting on an object.



Find the magnitude of the resultant force in Newton and its direction to the nearest degree.

3

c. Find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 x \, dx$

3

d. Find the volume, in terms of π of the solid formed when the area bounded by the curve $y = \frac{1}{2} \log_e x$, the lines $y = -1$ and $y = 2$ and the y -axis is rotated about the y -axis.

3

e. Use the compound angles identities to first simplify and then solve $\cos 3x - \cos 2x + \cos x = 0$ for $0 \leq \theta \leq \pi$.

3

Question 14**Start on a new answer booklet****15 Marks**

- a. Use mathematical induction to prove that, for any integer $n \geq 1$

$3^{2n+6} - 4^{n+1}$ is divisible by 5.

4

b. i. Prove that $\cot\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1 - \cos\theta}$

2

- ii. Hence find the exact value with a rationalised denominator of

2

$\cot\left(\frac{\theta}{2}\right)$ given that $\sin\theta = \frac{5}{6}$ and $\frac{\pi}{2} < \theta < \pi$.

- c. A cone is expanding. At a time t seconds, it has a radius r cm.

A perpendicular height $h = r\sqrt{3}$ cm and its surface area is increasing

at a rate of $\left(\frac{\pi}{\sqrt{3}}\right)^{\frac{1}{3}} \text{ cm}^2 \text{ s}^{-1}$.

As the cone expands it remains conical and similar to its original shape.

- i. Show that its surface area is $S = 3\pi r^2 \text{ cm}^2$

and its volume $V = \frac{\sqrt{3}}{3} \pi r^3 \text{ cm}^3$.

2

- ii. Show that $\frac{dV}{dt} = \frac{\sqrt{3}}{6} V^{\frac{1}{3}}$.

3

- iii. Given that the initial volume of the cone is $27\,000 \text{ cm}^3$

calculate its volume when $t = 250\sqrt{3}$ seconds.

2**End of Task**

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Solutions.

Student's Name: Multiple Choice Teacher's Code: Q's 1-10

Question Number	Marker use only



Saint Ignatius' College
RIVERVIEW

Mathematics

Writing Booklet

(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	
(h)	

Instructions

- Start each question in a new booklet.
- Except for multiple choice answers, you should show relevant mathematical reasoning and/or calculations.
- You may ask for an extra Writing Booklet to answer this question if you need more space.
- If you do not attempt a question you must still hand in a Writing Booklet, with your Name, your Teacher's Name, the Question Number the words 'NOT ATTEMPTED' written clearly on the front cover.
- Write the number of each section part inside the margin at the beginning of each answer.
- Write using black or blue pen.
- Write on the ruled pages only.

You may not remove any Writing Booklets, used or unused, from the examination room.

Start your answer here.

Multiple Choice.

Comments.

1. $f(x) = \sqrt{x} - 3$

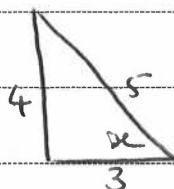
$D: x \geq 0$ $R: y \geq -3$

$\therefore f^{-1}(x)$ $D: x \geq -3$, $R: y \geq 0$

(A)

2. Value of $\sin 2x$

$\sin x = 0.8 = \frac{4}{5}$



$\sin 2x = 2 \sin x \cos x$

$= 2 \times \frac{4}{5} \times \left(\frac{-3}{5}\right) = -\frac{24}{5}$

x is obtuse $\therefore \sin 2x = -\frac{24}{5}$

(B)

3. $y = \cos^{-1}(2x)$

$y = \cos^{-1}\left(\frac{2x}{4}\right) + 3$

(C)

4. $y = f(x) + 3$

Moving all pts. up 3 (+3)

resulting in 3 roots, which is

2 more than the original fn.

(B)

$-2 \times f(x)$

1 Soln.

$f^{-1}(x)$

1 Soln.

$f(x+3)$

1 Soln.

$$5. f(x) = \frac{3+e^{2x}}{5} \quad f(x) > \frac{5}{3}$$

$$y = \frac{3+e^{2x}}{5}$$

$$\therefore x = \frac{3+e^{2y}}{5}$$

$$5x-3 = e^{2y}$$

$$\ln|5x-3| = 2y$$

$$y = \frac{1}{2} \ln|5x-3|$$

①

6. x-intercept will occur when $y=0$

$$\therefore \cos 2t = 0$$

$$\therefore 2t = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$t = \frac{\pi}{4}, \frac{3\pi}{4}, \dots$$

$\therefore$ 1st x-intercept at $t = \frac{\pi}{4}$

$$\text{Sub. } t = \frac{\pi}{4}, \text{ in } x = -\ln t$$

$$x = -\ln\left(\frac{\pi}{4}\right)$$

$$\approx 0.24 \text{ (2dp)}$$

①

$$7. a = \sqrt{pq}$$

$$= \sqrt{0.27 \times 0.73}$$

$$\approx 0.4440$$

①

$$8. \quad 3x + 2y + 1 = 0 \quad (L)$$

$$2y = -3x - 1$$

$$y = -\frac{3}{2}x - \frac{1}{2}$$

$$\therefore m_L = -\frac{3}{2} \quad \therefore m_{\perp} = \frac{2}{3}$$

$$\therefore \underline{3x + 2y}$$

(A)

$$9. \quad \frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}}$$

$$\text{Let } \tan \frac{\theta}{2} = t$$

$$\therefore \cot \frac{\theta}{2} = \frac{1}{t}$$

$$\frac{\frac{1}{t} - t}{\frac{1}{t} + t} = \frac{1 - t^2}{t} \times \frac{t}{1 + t^2}$$

$$= \frac{1 - t^2}{1 + t^2} = \cos \theta$$

(A)

10.

(A)

Student's Name: Question 11.

Teacher's Code: _____

Question Number	Marker use only
11	



Saint Ignatius' College
RIVERVIEW

Mathematics

Writing Booklet

(a)	
(b)	
(c)	
(d)	
(e)	
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Question 11

a. ~~$(2x-1)^3$~~

a. $P(x) = (x-2)(x+1)Q(x) + (x-b)$

$$P(-1) = 3$$

$$\therefore P(-1) = (-1-2)(-1+1)Q(x) + (-1-b) = 3$$

$$= (-3)(0)Q(x) + (-1-b) = 3$$

$$-1-b = 3$$

$$-b = 4$$

$$\therefore b = -4$$

✓ since $P(x) \div (x+1)$ Rem = 3

many didn't recognize this

✓ evaluating correctly

b. (i) $\frac{dy}{dt} = 1-y$

$$\int dt = \int \frac{dy}{1-y}$$

or $t = \int \frac{1}{1-y} dy$

✓ showing this point or equivalent

$$t = -\ln|1-y| + A$$

$$-\ln|1-y| = t - A \quad \text{or} \quad \ln|1-y| = A - t$$

✓ integrating

(ii) When $t=0$, $y=0$ $\therefore \ln|1| = A$

$$\text{i.e. } A = 0$$

Since $0 \leq y \leq 1$ $\ln(1-y) = -t$

$$1-y = e^{-t}$$

$$y = 1 - e^{-t}$$

✓ finding the equation

(iii) When $t=3$, $y = 1 - \frac{1}{e^3}$

Velocity vector when $t=3$

$$\vec{r} = \frac{1}{y} \underline{i} + (1-y) \underline{j} \quad \text{Given.}$$

$$y = 1 - \frac{1}{e^3} = \frac{e^3 - 1}{e^3}$$

$$\therefore \frac{1}{y} = \frac{e^3}{e^3 - 1}$$

$$1 - y = \frac{1}{e^3}$$

Sub. the values in $\vec{r}$

$$\therefore \vec{r} = \frac{e^3}{e^3 - 1} \underline{i} + \frac{1}{e^3} \underline{j}$$

also allowed

$$\vec{r} = \frac{1}{1 - e^{-3}} \underline{i} + e^{-3} \underline{j}$$

$$c. \int_0^1 \frac{e^x}{(1+e^x)^2} dx$$

$$u = e^x + 1$$

$$du = e^x dx$$

$$@ x=0, u=2$$

$$@ x=1, u=e+1$$

$$\therefore \int_2^{e+1} \frac{du}{u^2}$$

✓ finding appropriate integral

$$\int_2^{e+1} u^{-2} du = \left[-u^{-1} \right]_2^{e+1}$$

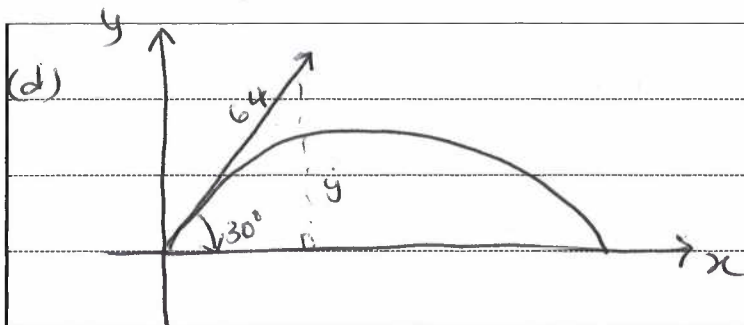
✓ integrating

$$= - \left[\frac{1}{u} \right]_2^{e+1} = - \left[\frac{1}{e+1} - \frac{1}{2} \right]$$

$$= - \left[\frac{2 - e - 1}{2(e+1)} \right] = - \left[\frac{1 - e}{2(e+1)} \right]$$

$$= \frac{e - 1}{2(e+1)} \quad (\text{In single fraction form})$$

✓ finding exact value as a single fraction.



$$(i) a_x = \ddot{x} = 0$$

$$v_x = \dot{x} = C_1$$

$$\text{At } t=0, v_x = 64 \cos 30^\circ = 32\sqrt{3}$$

$$\therefore \dot{x} = 32\sqrt{3} \text{ ms}^{-1}$$

$$x = 32\sqrt{3}t + C_2$$

$$\text{When } t=0, x=0 \quad \therefore C_2=0$$

$$\therefore x = 32\sqrt{3}t$$

✓
showing initial $\dot{x}$
and evaluating constants

$$(ii) \text{Vertically } a_y = \ddot{y} = -10$$

$$v_y = \dot{y} = -10t + C_3$$

$$\text{at } t=0, v_y = 64 \sin 30^\circ = 32 \text{ ms}^{-1} = C_3$$

$$\text{i.e. } \dot{y} = -10t + 32$$

$$\therefore y = -5t^2 + 32t + C_4$$

$$\text{at } t=0, y=0 \quad \therefore C_4=0$$

$$y = -5t^2 + 32t$$

✓
showing initial $\dot{y}$
and evaluating constants

✓
integrating and
evaluating constants

$$(iii) x = 32\sqrt{3}t$$

$$t = \frac{x}{32\sqrt{3}}$$

✓ making t subject

$$y = -5t^2 + 32t$$

$$= -5 \left[\frac{x}{32\sqrt{3}} \right]^2 + 32 \left[\frac{x}{32\sqrt{3}} \right]$$

$$y = \frac{-5x^2}{3072} + \frac{\sqrt{3}x}{3}$$

✓ finding cartesian equation
(many students did not know what this means)

You may ask for an extra Writing Booklet if you need more space to answer this question.

QUESTION 12

$$a) \frac{x-5}{2x+1} \geq 1 \quad x \neq -\frac{1}{2}$$

$$\frac{x-5}{2x+1} \times (2x+1)^2 \geq 1 \times (2x+1)^2$$

$$(x-5)(2x+1) \geq (2x+1)^2$$

$$(2x+1)^2 - (x-5)(2x+1) \leq 0$$

$$(2x+1)(x+6) \leq 0$$



$$-6 \leq x < -\frac{1}{2}$$

← 1mk

Too many did not spot that $x \neq -\frac{1}{2}$.

1mk for each inequality.

$$b) i) y = e^{2x} + e^{-2x}$$

$$y' = 2e^{2x} - 2e^{-2x}$$

$$y'' = 4e^{2x} + 4e^{-2x}$$

Well done

← 1mk

$$ii) y'' + ay' + by = 5e^{2x} + e^{-2x}$$

$$\begin{aligned} \text{RHS} &= 4e^{2x} + 4e^{-2x} + a(2e^{2x} - 2e^{-2x}) + b(e^{2x} + e^{-2x}) \\ &= 4e^{2x} + 4e^{-2x} + 2ae^{2x} - 2ae^{-2x} + be^{2x} + be^{-2x} \\ &= e^{2x}(4 + 2a + b) + e^{-2x}(4 - 2a + b) \end{aligned}$$

1mk for correct setting out

$$\begin{aligned} \therefore 5 &= 4 + 2a + b & 1 &= 4 - 2a + b \\ 2a + b &= 1 \text{ --- ①} & 2a - b &= 3 \text{ --- ②} \\ \text{①} + \text{②} & 4a = 4 & 2 - b &= 3 \\ & a = 1 & b &= -1 \end{aligned}$$

Imk each
for
 $a=1$
 $b=-1$

c) 5 Blue 6 Black 3 Red.

Poorly done

$$\begin{aligned} \text{Exactly 2 Blue in 3} &= {}^5C_2 \times {}^9C_1 \\ &= 90 \end{aligned}$$

$$\begin{aligned} \text{Arrangements of 3 from 14} &= {}^{14}C_3 \\ &= 364 \end{aligned}$$

Imk each

$$\therefore P(e) = \frac{90}{364} \text{ OR } \frac{45}{182}$$

OR

OR CORRECT FROM TREE DIAGRAM

$$d) i) \int \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx$$

$$u = 1 - x^2 \quad = -\frac{1}{2} \int \frac{1}{\sqrt{u}} du$$

$$\frac{du}{dx} = -2x \quad = -\frac{1}{2} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right] + C$$

$$du = -2x dx \quad = -\sqrt{u} + C$$

$$= -\sqrt{1-x^2} + C$$

Imk for
set up

Imk

$$\begin{aligned} \text{ii) } \frac{d}{dx} x \cos^{-1} x &= x \cdot \frac{-1}{\sqrt{1-x^2}} + \cos^{-1} x \\ &= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} \end{aligned}$$

link for each "part"

$$\text{iii) } \int \cos^{-1} x \, dx = ???$$

From (ii)

$$\frac{d}{dx} x \cos^{-1} x = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

$$\cos^{-1} x = \frac{d}{dx} x \cos^{-1} x + \frac{x}{\sqrt{1-x^2}}$$

$$\begin{aligned} \therefore \int \cos^{-1} x &= x \cos^{-1} x + \int \frac{x}{\sqrt{1-x^2}} \, dx \\ &= x \cos^{-1} x - \sqrt{1-x^2} + C \end{aligned}$$

Well done
by most

link for each
"part"

Start your answer here.

Question 13

a. $u = 2\hat{i} + 3\hat{j}$ $v = \hat{i} + 2\hat{j}$

* Well answered

(i) $u - v = 2\hat{i} - \hat{i} + 3\hat{j} - 2\hat{j}$
 $= \hat{i} + \hat{j}$

✓

(ii) $u - \lambda v = -2\hat{i} - 5\hat{j}$

$\therefore 2\hat{i} + 3\hat{j} - \lambda\hat{i} - \lambda 2\hat{j} = -2\hat{i} - 5\hat{j}$

$(2 - \lambda)\hat{i} + (3 - 2\lambda)\hat{j} = -2\hat{i} - 5\hat{j}$

$\therefore 2 - \lambda = -2$

check

$\lambda = 4$

$3 - 2\lambda = -5$

$-2\lambda = -8$

$\lambda = 4$

✓ * Well answered

$\therefore \lambda = 4$

(iii) $au + bv = \hat{i}$

$a(2\hat{i} + 3\hat{j}) + b(\hat{i} + 2\hat{j}) = \hat{i} + 0\hat{j}$

$2a\hat{i} + b\hat{i} + 3a\hat{j} + 2b\hat{j} = \hat{i} + 0\hat{j}$

$2a + b = 1$

$3a + 2b = 0$

$b = 1 - 2a$

$3a + 2(1 - 2a) = 0$

$3a + 2 - 4a = 0$

$-a = -2$

$\therefore b = 1 - 2(2)$

$a = 2$

$b = -3$

$\therefore a = 2, b = -3$

✓ * well answered.

b. Method: write each vector in the form of $a\mathbf{i} + b\mathbf{j}$ and then add for Resultant Force.

$$40\text{ N} : 40 \cos 150^\circ \mathbf{i} + 40 \sin 150^\circ \mathbf{j} = -20\sqrt{3}\mathbf{i} + 20\mathbf{j}$$

$$30\text{ N} : 30 \cos 210^\circ \mathbf{i} + 30 \sin 210^\circ \mathbf{j} = -15\sqrt{3}\mathbf{i} - 15\mathbf{j}$$

$$20\text{ N} : 20 \cos (-30^\circ) \mathbf{i} + 20 \sin (-30^\circ) \mathbf{j} = 10\sqrt{3}\mathbf{i} - 10\mathbf{j}$$

$$\mathbf{R} = (-20\sqrt{3} - 15\sqrt{3} + 10\sqrt{3})\mathbf{i} + (20 - 15 - 10)\mathbf{j}$$

$$= -25\sqrt{3}\mathbf{i} - 5\mathbf{j}$$

Magnitude:

$$R : \sqrt{(-25\sqrt{3})^2 + (-5)^2} = \sqrt{1900} \\ = 10\sqrt{19} \text{ N}$$

$$\text{Direction: } \tan \theta = \frac{-5}{-25\sqrt{3}} = \frac{1}{5\sqrt{3}}$$

$$\therefore \theta = \tan^{-1}\left(\frac{1}{5\sqrt{3}}\right) = 186^\circ 35' \\ \doteq 187^\circ$$

Hence, the resultant vector has magnitude $10\sqrt{19} \text{ N}$ and forms an angle of 187° with the positive direction of the x -axis.

$$c. \int_{\pi/4}^{\pi/2} \sin^2 x dx = \frac{1}{2} \int_{\pi/4}^{\pi/2} 1 - \cos 2x dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_{\pi/4}^{\pi/2} = \frac{1}{2} \left[\frac{\pi}{2} - \frac{1}{2} \sin \pi \right] - \left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right)$$

$$= \frac{1}{2} \left[\frac{\pi}{2} - 0 - \frac{\pi}{4} + \frac{1}{2} \right] = \frac{1}{2} \left[\frac{\pi}{4} + \frac{1}{2} \right]$$

$$\text{OR} = \frac{\pi}{8} + \frac{1}{4}$$

* Some students

adopted a visual, geometric approach and made mistakes.

* mistakes were made

when calculating various sin and cos values.

✓

* Follow on marks

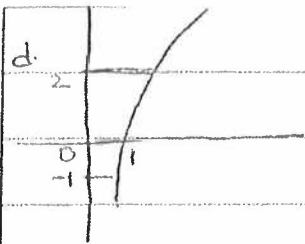
are awarded if you used correct method to

calculate length and value of angle.

* Most students adopted correct approach but some

made silly errors with signs and when subtracting

Additional writing space on back page.



$$V = \pi \int_a^b x^2 dy$$

$$= \pi \int_{-1}^2 (e^{2y})^2 dy$$

$$= \pi \int_{-1}^2 e^{4y} dy = \frac{\pi}{4} [e^{4y}]_{-1}^2$$

$$= \frac{\pi}{4} (e^8 - e^{-4}) u^3$$

$$y = \frac{1}{2} \log_e x$$

$$2y = \log_e x$$

$$e^{2y} = x$$

* Well answered

* Some students tried to calculate the answer using $V = \pi \int y^2 dx$ which is incorrect.

$$(e) \cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$\therefore \cos 3x + \cos x = 2 \cos \left(\frac{3x+x}{2} \right) \cos \left(\frac{3x-x}{2} \right)$$

$$= 2 \cos 2x \cos x$$

$$\therefore \cos 3x + \cos x - \cos 2x = 0$$

$$2 \cos 2x \cos x - \cos 2x = 0$$

$$\cos 2x (2 \cos x - 1) = 0$$

$$\cos 2x = 0$$

$$\cos x = \frac{1}{2}$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = \frac{\pi}{3}$$

1st 2 Quad. only.

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\text{Solu: } x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{3}$$

* If you used identities which were relevant to try and prove the statement but did not finish

proof you received 1 mark.

You may ask for an extra Writing Booklet if you need more space to answer this question.

14a

R.T.P. $3^{2n+6} - 4^{n+1} / 5 \quad n \geq 1$

let $n=1$

$$\begin{aligned} \text{LHS} &= 3^{2(1)+6} - 4^{1+1} \\ &= 3^8 - 4^2 \\ &= 6561 - 16 \\ &= 6545 \\ &= 5 \times 1309 \quad \checkmark \end{aligned}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

i.e. $3^{2k+6} - 4^{k+1} = 5M$, M an integer

OR

$$3^{2k+6} = 5M + 4^{k+1}$$

let $n=k+1$

R.T.P. $3^{2(k+1)+6} - 4^{(k+1)+1} = 3^{2k+8} - 4^{k+2} = 5N$, N an integer.

$$\text{LHS} = 3^{2k+8} - 4^{k+2}$$

$$= 3^2 \cdot 3^{2k+6} - 4 \cdot 4^{k+1}$$

$$= 9 \times 3^{2k+6} - 4 \cdot 4^{k+1} \quad \checkmark$$

$$= 9(5M + 4^{k+1}) - 4 \cdot 4^{k+1} \quad \text{from assumption} \quad \checkmark$$

$$= 45M + 9 \times 4^{k+1} - 4 \cdot 4^{k+1}$$

$$= 45M + 5 \times 4^{k+1}$$

$$= 5(9M + 4^{k+1})$$

$$= 5N, \quad N \text{ an integer} \quad \checkmark$$

$\therefore$ True by principle of Mathematical Induction

b) i.) Prove $\cot\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1-\cos\theta}$

$$\begin{aligned}
 \text{RHS} &= \frac{\sin\left(2\left(\frac{\theta}{2}\right)\right)}{1-\cos\left(2\left(\frac{\theta}{2}\right)\right)} \\
 &= \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{1-(1-2\sin^2\frac{\theta}{2})} \checkmark \\
 &= \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} \checkmark \\
 &= \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}} \\
 &= \cot\frac{\theta}{2}
 \end{aligned}$$

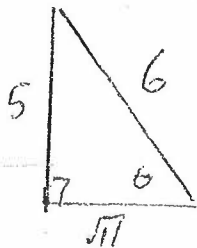
(OR)

Let $\tan\frac{\theta}{2} = t \therefore \cot\left(\frac{\theta}{2}\right) = \frac{1}{t}$

$\sin\theta = \frac{2t}{1+t^2}$ $\cos\theta = \frac{1-t^2}{1+t^2}$

$$\begin{aligned}
 \text{LHS} &= \frac{2t}{1+t^2} \checkmark \\
 &= \frac{2t}{1-\frac{1-t^2}{1+t^2}} \checkmark \\
 &= \frac{2t}{1+t^2-1+t^2} \\
 &= \frac{2t}{2t^2} \\
 &= \frac{1}{t}
 \end{aligned}$$

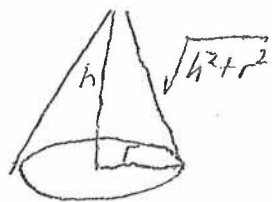
(ii) | $\sin\theta = \frac{5}{6}$ $\frac{\pi}{2} \leq \theta \leq \pi$



$\therefore \cot\left(\frac{\theta}{2}\right) = \frac{\frac{5}{6}}{1-\left(-\frac{\sqrt{11}}{6}\right)} \checkmark$

$$\begin{aligned}
 &= \frac{5}{6+\sqrt{11}} \times \frac{6-\sqrt{11}}{6-\sqrt{11}} \\
 &= \frac{30-5\sqrt{11}}{36-11} \\
 &= \frac{30-5\sqrt{11}}{25} = \frac{6-\sqrt{11}}{5}
 \end{aligned}$$

c.)



$$h = r\sqrt{3}$$

$$\frac{dA}{dt} = \left(\frac{\pi}{3}\right)^{1/3}$$

$$(i) S = \pi r l + \pi r^2$$

$$l = \sqrt{h^2 + r^2} = \sqrt{3r^2 + r^2} = 2r \quad \checkmark$$

$$\therefore S = \pi r (2r) + \pi r^2$$

$$= 2\pi r^2 + \pi r^2 = 3\pi r^2$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi r^2 r \sqrt{3} = \frac{\sqrt{3}}{3} \pi r^3 \quad \checkmark$$

$$(ii) \frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$$

$$= \frac{dV}{dr} \times \frac{dr}{dA} \times \frac{dA}{dt}$$

$$= \sqrt{3} \pi r^2 \times \frac{1}{6\pi r} \times \left(\frac{\pi}{3}\right)^{1/3} \quad \left. \vphantom{\frac{dV}{dr}} \right\}^{\frac{1}{3}}$$

$$= \frac{\sqrt{3}}{6} \times r \times \left(\frac{\pi}{3}\right)^{1/3} \quad \checkmark$$

$$= \frac{\sqrt{3}}{6} \times \left(\frac{\sqrt{3}}{3} \pi r^3\right)^{1/3} = \frac{\sqrt{3}}{6} V^{1/3}$$

$$(iii) t=0 \quad V=27,000$$

$$\frac{dV}{dt} = \frac{\sqrt{3}}{6} V^{1/3}$$

$$V^{-1/3} dV = \frac{\sqrt{3}}{6} dt$$

$$3 \frac{V^{2/3}}{2} = \frac{\sqrt{3}}{6} t + C$$

$$\therefore C = 3 \left(\frac{27,000}{2} \right)^{2/3} = 135$$

$$V = \left(\frac{\sqrt{3}}{9} t + 900 \right)^{3/2} \quad \checkmark$$

$$t=250\sqrt{3} \quad V=30835.51 \quad \checkmark$$